

MTH304 - Midsem 1

1. In S_5 , consider the permutations

$$\sigma := (13524) \quad \text{and} \quad \tau = (14253)$$

i.e. $\sigma(1) = 3, \sigma(3) = 5, \dots$. Find an element $x \in S_5$ such that

$$\sigma x = \tau$$

2. Show that the multiplicative subgroup

$$\{\pm 1, \pm i, \pm j, \pm k\}$$

of the group of Quaternions is non-Abelian, but all its proper subgroups are Abelian. Are all proper subgroups normal?

3. Let G be an Abelian group. Show that all elements of G of finite order form a subgroup of G .
4. Show that the residue classes of $1, 2, 4, 5, 7, 8$ modulo 9 form a group under multiplication, which is isomorphic to the cyclic group $\mathbb{Z}/6\mathbb{Z}$.
5. Let $(\langle \mathbb{Z}/13\mathbb{Z} \rangle^\times, \times)$ denote the group of units in $\mathbb{Z}/13\mathbb{Z}$. Find an element of order 4 and another of order 3 in this group.